NATIONAL UNIVERSITY BANGLADESH

PROVISIONAL

Syllabus
Bachelor of Science (Honours) Second Year
Department of Environment Science

Effective from the Academic Year: 2024–2025 and onward

1. Vision

We expect that on the hundredth anniversary of the National University, our commitment to "Progress for Everyone" will be evident in all areas of our operations and stakeholder engagements.

We leverage our core strength in education, research, and community engagement to address pressing global issues and foster sustainable development.

By 2042, National University aims to strengthen its positions as Bangladesh's top academic institution and rank among the world's leading universities. Our focus will be on expanding expertise building strategic partnerships, and increasing our global influence. Our services will be measured by the achievements of the individuals and communities we serve.

2. **Mission**

To promote "Progress for Everyone" by fostering transformative education, conducting pioneering research, and encouraging meaningful collaboration with individuals, communities, and partners both in Bangladesh and around the world.

3. Name of the Program: B.Sc. (Honours) in Environment Science

4. Program Educational Objectives (PEOs)

PEO 1: To produce graduates with comprehensive scientific knowledge, practical competence and interdisciplinary perspective to address contemporary and emerging environment challenges in local, national and global context.

PEO 2: To develop graduates capable of conducting rigorous, interdisciplinary and evidence-based research and applying innovation approaches, technologies and data driven methods to environmental monitoring, assessment, management and problem solving.

PEO 3: To develop graduates who can contribute to the generation of reliable, timely, and relevant environmental data and solutions to support research, planning, and decision-making processes—especially in alignment with the Sustainable Development Goals (SDGs) of Bangladesh.

PEO 4: To develop environmentally responsible professionals who demonstrate leadership, effective communication, collaboration, ethical judgment, cultural awareness and a commitment to lifelong learning and professional development.

PROVISIONAL

5. Program Learning Outcomes (PLOs):

PLO 1: Environmental Knowledge
Understand and apply core concepts of environmental science—including ecology, climate, biodiversity, pollution, and natural resource management to solve real-life, academic, and professional environmental problems.

PLO 2: Problem-Solving Ability
Recognize, formulate, and solve environmental problems using scientific reasoning, field data, models, and relevant analytical techniques to draw logical and evidence-based solutions.

PLO 3: Sustainable Solutions
Design and conduct scientific studies to address environmental challenges that meet the specified needs of sustainable living, human welfare, public health, and ecological conservation.

PLO 4: Research Skills
Collect, analyze, and interpret environmental data using qualitative and quantitative methods; present results with clarity and draw appropriate scientific methods.

PLO 5: Use of Modern Tools
Utilize modern tools and technologies such as GIS, remote sensing, environmental sensors, and simulation models relevant to environmental monitoring and assessment.

PLO 6: Environment and Society
Apply environmental science knowledge to address issues in public health, sustainable development, natural resource use, disaster risk reduction, and community well-being.

PLO 7: Ethics and Responsibility
Develop and uphold strong ethical principles and environmental responsibility in personal and professional contexts for the betterment of society and the planet.

PLO 8: Teamwork and Leadership Skills
Communicate effectively and work collaboratively in multidisciplinary teams; demonstrate leadership skills to address and manage complex environmental issues.

PLO 9: Project and Time Management
Plan, manage, and execute environmental fieldwork, laboratory analysis, and research projects efficiently, both individuals and groups, demonstrating responsibility and timely action.

PLO 10: Lifelong Learning and career opportunities

Engage in continuous learning to keep pace with evolving environmental challenges, scientific advancements, technologies, and policies for personal and professional development. Demonstrate preparedness for careers in environmental consulting, research institutions, teaching, governmental and non-governmental organizations.

PROVISIONAL

6. Mapping PEO with PLO:

	PEO-1	PEO-2	PEO-3	PEO-4
PLO-1	✓	✓		
PLO-2	✓	✓		
PLO-3	✓	✓	✓	
PLO-4	✓	✓		
PLO-5	✓	✓	✓	
PLO-6	✓		✓	✓
PLO-7			✓	✓
PLO-8			✓	✓
PLO-9	✓	✓		✓
PLO-10	✓	✓	✓	✓

PROVISIONAL

7. Courses, Marks and Credit Distribution

Second Year

Course Code	Course Title	Marks	Credits
	Core Courses		
8224451	Environmental Earth and Soil Science	100	4
8224453	Environmental Chemistry	100	4
8224455	Environmental Biology	100	4
8224457	Atmospheric Science	100	4
8224460	Laboratory and Field Work on Environmental Sciences	100	4
	Allied Courses **Any two of the following courses**		
8223255	Physical Geography and Environment –Bangladesh perspective	100	4
8224456	Environment and Economics	100	4
8224457	Environmental Biochemistry	100	4
	Compulsory Courses		
8221171	Functional English	100	4
8226601	Applied Information and Communication Technology (Applied ICT)	75	3
8226602	Practical: Applied ICT	25	1
8224464	Viva-voce (Environmental Science)	50	2
	Total =	**950**	**38**

PROVISIONAL

Course Code	8224451	Marks: 100	Credits: 4	Class Hours:56
Course Title	Environmental Earth and Soil Science			

Course Description

This course introduces students to the fundamental principal of Environmental Earth Sciences with comparison hazardous Earth processes, environmental health, environmental effects on mining, and soil science. It examines natural hazards, geological factors affecting human health, environmental consequences of mining, and the physical, chemical, and biological properties of soil. The course also develops basic skills in soil observation, characterization, and sampling, with emphasis on environmental sustainability and agricultural applications.

Course Objectives

This course will expose students to

1. understand an introductory idea about the hazardous earth processes and the geological aspects of environmental health and as emerging field.
2. understand the environmental effects of mining.
3. provide a fundamental idea on physical, chemical and microbiological aspects of soil.
4. focus on the basic foundation of studying a complex dynamic system like soil.

Course Learning Outcomes (CLOs)

CLO1	familiar with the fundamental principles of an emerging subject
CLO2	understand the importance of the subject and extent understand and application the importance
CLO3	get idea about the composition of soil and its relationship with plant growth, evaluate and recommend appropriate soil management for sustainable agricultural production.
CLO4	understand the geological aspects of environmental health.
CLO5	identify, characterize, and explain hazardous earth processes from local to global.
CLO6	Identity and perform basic soil sampling using standard scientific methods.

Mapping with CLOs with PLOs

CLO / PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	✓									
CLO2	✓	✓	✓			✓				
CLO3	✓	✓	✓			✓				
CLO4	✓	✓	✓			✓				
CLO5	✓	✓	✓			✓				
CLO6	✓	✓		✓	✓				✓	

PROVISIONAL

Course Content

Module No.	Content	Teaching–Learning Strategies	Assessment Strategies	CLOs
1	**Philosophy and Fundamental Principles of Environmental Earth Sciences:** Cultural basis of environmental awareness; fundamental concepts of environmental geology; Earth materials and processes.	Lecture (PPT), short videos, group discussion, Q&A	Class participation, short quiz, formative assessment	CLO1
2	**Hazardous Earth Processes:** Natural hazards; disaster prediction and risk assessment; human responses to hazards; river flooding, earthquakes, mass wasting, volcanic activity, coastal and wind hazards their causes, types, impacts and prediction.	Lecture (PPT), videos, group discussion, case-based learning, Q&A	Quiz, case-study analysis, class presentation, written assignment	CLO1, CLO2, CLO5
3	**Geological Aspects of Environmental Health:** Health factors; geological controls on environmental health; trace elements and health; chronic diseases and geological environments; asbestos and radon hazards.	Lecture (PPT), videos, group discussion, case-based learning, Q&A	Quiz, case-study analysis, assignment, class discussion	CLO2, CLO4, CLO5
4	**Environmental Effects of Mining:** Mining, ores, protore, metals and metalloids; acid mine drainage; pit lakes; pollution treatment; primary, secondary and tertiary impacts; pre-mining and post-mining contamination.	Lecture (PPT), videos, group discussion, case-based learning	Written assignment, case-study report, presentation, quiz	CLO2, CLO4, CLO5
5	**Concepts of Soil:** Soil as an ecosystem component; importance of soil; soil as a medium for plant growth and as a source and sink of pollutants; soil minerals; global distribution of soils.	Lecture (PPT), concept mapping, Q&A	Quiz, concept map, short assignment, class participation	CLO1, CLO2, CLO3, CLO6
6	**Soil Formation, Profile, Texture and Color:** Factors of soil formation; weathering of rocks and minerals; types and controls of weathering; master horizons; soil particles and classification;	Lecture (PPT), concept mapping,	Practical exercise, field observation report, quiz, assignment	CLO1, CLO3, CLO6

PROVISIONAL

	textural classes and their significance; soil color.	field-based observation		
7	**Soil Structure:** Genesis and classification of soil structure; relationship of soil structure with air and water; particle density, bulk density and soil porosity.	Lecture (PPT), concept mapping, field-based observation	Practical exercise, numerical/problem-solving task, field report, quiz	CLO1, CLO3, CLO6
8	**Soil Water and Soil Air:** Physical classification of soil water; forces of water retention; soil-water constants; plant-available water; effects of excess water; infiltration, runoff and permeability; composition and importance of soil air.	Lecture (PPT), concept mapping, Q&A	Quiz, problem-solving exercise, assignment, class discussion	CLO1, CLO3
9	**Soil Biota:** Plant roots, microflora, mesofauna and macrofauna; earthworms, nematodes, arthropods, protozoa, bacteria and viruses; ecological functions of soil biota.	Lecture (PPT), concept mapping, field-based observation	Quiz, field observation, assignment, presentation	CLO1, CLO3

PROVISIONAL

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Brady, N. C., & Weil, R. R. (2017). *The nature and properties of soils* (15th ed.). Pearson.

Hillel, D. (1980). *Fundamentals of soil physics*. Academic Press.

Keller, E. A. (2010). *Environmental geology* (9th ed.). Pearson.

Killham, K. (1994). *Soil ecology*. Cambridge University Press.

Sparks, B. W. (1986). *Geomorphology* (3rd ed.). Longman.

Tan, K. H. (2011). *Principles of soil chemistry* (4th ed.). CRC Press.

Thornbury, W. D. (1969). *Principles of geomorphology* (2nd ed.). John Wiley & Sons

PROVISIONAL

Course Code	8224453	Marks: 100	Credits: 4	Class Hours:56
Course Title	Environmental Chemistry			

Course Description

This course introduces the fundamental principles of environmental chemistry and their application to aquatic, atmospheric, and soil environments. It examines biogeochemical cycles, environmental pollutants, their behavior and impacts, and scientifically sound approaches to pollution control, environmental protection, and sustainable development.

Course Objectives

The objectives of this course are to:

1. explain the fundamental concepts, principles, scope, and interdisciplinary nature of environmental chemistry and its relationship with other branches of chemical sciences.
2. develop an understanding of the chemical composition and processes occurring in the atmosphere, hydrosphere, lithosphere, and biosphere, with emphasis on environmental sustainability.
3. explain the chemical behavior, transport, transformation, and environmental impacts of pollutants in air, water, and soil.
4. develop knowledge of aquatic chemistry, atmospheric chemistry, soil chemistry, and biogeochemical cycles governing environmental quality.
5. develop analytical and problem-solving skills to evaluate environmental pollution and recommend appropriate pollution prevention and management strategies.

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO	Course Learning Outcome
CLO1	Explain the fundamental concepts, principles, scope, and interdisciplinary nature of environmental chemistry.
CLO2	Analyze the chemical processes governing aquatic, atmospheric, and soil environments using environmental chemistry principles.
CLO3	Evaluate the characteristics, behavior, transport, and transformation of environmental pollutants in different environmental compartments.
CLO4	Assess the significance of biogeochemical cycles and their influence on environmental sustainability.
CLO5	Analyze environmental pollution problems and recommend scientifically sound pollution control and management strategies.

PROVISIONAL

Mapping of CLOs with PLOs

CLO/PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	✓									✓
CLO2	✓	✓		✓	✓	✓				✓
CLO3	✓	✓	✓	✓	✓	✓	✓			✓
CLO4	✓		✓	✓		✓	✓			✓
CLO5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Course Contents

Module No.	Content	Teaching–Learning Strategies	Assessment Strategies	CLOs
1	**Environmental Chemistry:** Definition, scope, importance of environmental chemistry; relationship with inorganic chemistry, organic chemistry, biochemistry, geochemistry, and biogeochemistry; role of environmental chemistry in sustainable development.	Lecture (PPT), concept mapping, multimedia presentation, interactive discussion, Q&A	Short quiz, concept-mapping exercise, class participation, short assignment	CLO1
2	**Aquatic Chemistry:** Introduction to aquatic chemistry; properties of water; major inorganic and organic constituents; dissolved gases; DO, COD, BOD; acidity, alkalinity, electrical conductivity; chemical equilibrium in aquatic systems; redox reactions; water quality standards.	Interactive lecture, numerical problem-solving, laboratory demonstration, case study, group discussion	Quiz, numerical/problem-solving exercises, laboratory assessment, case-study analysis	CLO2, CLO3
3	**Atmospheric Chemistry:** Concepts of atmospheric chemistry; atmospheric composition; global, regional and local atmospheric processes; atmospheric carbon dioxide; oxygen and nitrogen chemistry; ambient and indoor air quality standards.	Multimedia lecture, climate data interpretation, animation, case study, collaborative learning	Quiz, data interpretation exercise, case-study analysis, class presentation	CLO2, CLO5
4	**Soil Chemistry:** Chemical nature and properties of soil; soil nutrients; soil pH; cation exchange capacity (CEC); oxidation-reduction processes in soil; chemistry of waterlogged soils; chemical transformations during drying.	Interactive lecture, laboratory demonstration, soil sample analysis, field observation,	Practical/laboratory assessment, soil analysis report, field observation report, quiz	CLO2, CLO3, CLO4

PROVISIONAL

		group discussion		
5	**Biogeochemical Cycles:** Principles and types of biogeochemical cycles; nutrient cycling; carbon, nitrogen, phosphorus, and sulphur cycles; human interference and environmental consequences.	Lecture, concept mapping, seminar presentation, collaborative learning, case-based discussion	Concept map, seminar presentation, written assignment, case-study assessment	CLO3 CLO4, CLO5
6	**Environmental Pollution:** Concepts, types, and effects of environmental pollution; global and regional pollution issues; ozone; oxides of carbon, nitrogen and sulphur; heavy metals; radionuclides; particulate matter; aerosols; photochemical smog; methane and hydrocarbons; organic solvents; pesticides; PCBs; dioxins; pollution prevention and control strategies.	Case study, pollution data analysis, seminar presentation, multimedia lecture, problem-based learning	Pollution data analysis, case-study report, seminar presentation, problem-solving assignment, written examination	CLO3, CLO5, CLO6

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Baird, C. (2000). *Environmental chemistry* (2nd ed.). W. H. Freeman and Company.

Baird, C., & Cann, M. (2021). *Environmental chemistry* (6th ed.). W. H. Freeman.

Dara, S. S., & Mishra, D. D. (2006). *A textbook of environmental chemistry and pollution control*. S. Chand Publishing.

Harrison, R. M. (2015). *Pollution: Causes, effects and control* (6th ed.). Royal Society of Chemistry.

Manahan, S. E. (2010). *Environmental chemistry* (9th ed.). CRC Press.

McBride, M. B. (1994). *Environmental chemistry of soils*. Oxford University Press.

PROVISIONAL

Course Code	8224453	Marks: 100	Credits: 4	Class Hours: 56
Course Title	Environmental Biology			

Course Description

This course provides an integrated understanding of cellular processes, genetics, genetic engineering, ecology, evolution, biodiversity, and organismal adaptation. It examines biological communities and ecosystems, ecological interactions, succession, biomes, behavioral ecology, plant animal relationships, and the adaptations of organisms to diverse environmental conditions.

Course Objectives

This course will expose students to

1. Apply their knowledge to grow new plant from vegetative parts of the older plant.
2. A thorough knowledge will be developed to the student on the basic of the primary behavioral categories relating to reproduction, sustenance, development and survival of biological species.

Course Learning Outcomes (CLOs)

CLO1	Explain the structure and functions of cells, including cellular respiration, photosynthesis, DNA, RNA, protein synthesis, cell division, and heredity.
CLO2	Understand ecosystems and genetic engineering technology.
CLO3	Analyze biodiversity: Evaluate the factors that maintain or threaten species richness and genetic variety and plant animal interrelationship.
CLO4	Apply scientific methods: Use hypothesis testing, ecological data, and reasoning to study living systems.
CLO5	Interpret behavioral ecology concepts, including life-history strategies, parental care, social behavior, and evolutionary adaptations.
CLO6	Apply biological and ecological principles to analyze environmental issues and propose scientifically sound approaches for biodiversity conservation and sustainable ecosystem management.

Mapping with CLOs with PLOs

CLO/PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	✓									✓
CLO2	✓	✓					✓			✓
CLO3	✓	✓	✓	✓		✓	✓			✓
CLO4	✓	✓		✓	✓				✓	✓
CLO5	✓	✓						✓		✓
CLO6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

PROVISIONAL

Course Content

Module No	Content	Teaching Learning Strategies	Assessment Strategies	CLOs
1	**Structures and functions of life:** mitochondria and the process of cell respiration; Chlorophlast and the process of photosynthesis, Significance of photosynthesis; structures of DNA and RNA, protein synthesis; chromosomes and cell division; heredity	Lecture (PPT), short videos, group discussion, Q&A	Short quiz, class participation, written assignment, formative assessment	CLO1
2	**Genetic engineering and recombinant DNA technology:** tools and methods of recombinant DNA technology; enzymes involved in recombinant DNA technology; Improvement of useful microorganisms – mutations and selection of strains, protoplast fusion technology; uses of recombinant DNA technology	Lecture (PPT), short videos, group discussion, case-based learning, Q&A	Quiz, case-study analysis, written assignment, class presentation	CLO1, CLO2
3	**Biological communities and ecosystems;** characteristics of biological populations; Interactions between diverse populations-neutralism, commensalism, synergism, mutualism, competition, amensalism (antagonism), parasitism and predation; human population dynamics, study of populations; presentation of demographic data; community concept; structure of communities; global distribution of terrestrial communities; patterns of diversity and global diversity; concept of ecosystem, trophic levels and productivity; ; ecosystem stability	Lecture (PPT), short videos, group discussion, case-based learning, Q&A	Quiz, demographic data analysis, case-study assessment, group presentation	CLO2, CLO3, CLO4
4	**Succession and biomes;** vegetation changes; patterns of succession; human influence on succession; world's terrestrial biomes; wetland and freshwater biomes; coastal and marine biomes.	Lecture (PPT), short videos, group discussion, case-based learning, Q&A	Short quiz, case-study analysis, assignment, presentation	CLO3, CLO6
5	**Ecological genetics;** sources of variation; patterns of variation, reproductive system-obligate cross-fertilization, facultative cross-fertilization, self-fertilization, seed apomixes; vegetative reproduction; genetic consequences of different reproductive systems	Lecture (PPT), concept mapping, Q&A	Concept map, quiz, written assignment, class discussion	CLO2, CLO3, CLO6

PROVISIONAL

6	**Behavioral ecology and sociology;** optimization theory; growth versus reproduction; parental care; r and k species; breeding systems in plants: alternative strategies; games theory; living in groups, their advantages and disadvantages; the unit of selection and social behavior; human socio-biology.	Lecture (PPT), concept mapping, field-based observation	Field observation report, quiz, assignment, group discussion	CLO5, CLO6
7	**Plant-animal interrelationships:** pairwise co-evolution; diffuse co-evolution; angiosperm-pollinator relationships; introduced/ exotic species	Lecture (PPT), concept mapping, field-based observation	Field observation report, case-study analysis, presentation	CLO3, CLO5, CLO6
8	**Adaptation of Organism:** Adaptation of plant -Introduction and classification of plants on the basis of adaptation, Morphological, anatomical and physiological adaptive characteristics of hydrophytes, xerophytes and halophytes; Adaptation of animal- natural selection and behavior, visual adaptation to unfavorable environment.	Lecture (PPT), concept mapping, Q&A	Quiz, comparative assignment, case-study analysis, class presentation	CLO3 CLO5 CLO6

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Dubey, R. C. (1995). *A textbook of biotechnology* (2nd ed.). S. Chand & Company Ltd.

Atlas, R. M., & Bartha, R. (2000). *Microbial ecology: Fundamentals and applications* (4th ed.). Benjamin/Cummings.

Chapman, J. L., & Reiss, M. J. (1999). *Ecology: Principles and applications* (2nd ed.). Cambridge University Press.

McFarland, D. (1985). *Animal behaviour: Psychobiology, ethology and evolution*. ELBS.

PROVISIONAL

Course Code	8224457	Marks: 100	Credits: 4	Class Hours:56
Course Title	Atmospheric Science			

Course Description

This course introduces the fundamental principles of atmospheric science, weather, and climate. It examines the structure and composition of the atmosphere, solar radiation and heat balance, atmospheric moisture, clouds and precipitation, air pressure and wind systems, atmospheric circulation, monsoons, storms, and climate classification. The course also introduces ENSO, La Niña, and the basic concepts of climate change, with emphasis on weather observation, forecasting, and remote-sensing applications.

Course Objectives

This course aims to:

1. Introduce students to the fundamental concepts, structure, composition, and processes of the atmosphere.
2. Develop an understanding of the major elements and controls of weather and climate.
3. Explain atmospheric moisture, pressure, wind systems, circulation, clouds, precipitation, and major weather phenomena.
4. Develop students' ability to interpret climatic regions, atmospheric observations, and basic meteorological and climate data.
5. Develop an understanding of climate variability, climate change, and the interactions between atmospheric processes and human environment systems

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcomes
CLO1	Explain the fundamental concepts, structure, composition, and major processes of the atmosphere.
CLO2	Describe and interpret the major elements of weather, including temperature, humidity, pressure, wind, clouds, and precipitation.
CLO3	Analyze atmospheric circulation, monsoons, air masses, cyclones, anticyclones, thunderstorms, and other major weather phenomena.
CLO4	Compare and interpret the major climatic regions and classifications of the world and explain their controlling factors.
CLO5	Apply basic meteorological observations and climate data to interpret weather and atmospheric conditions.
CLO6	Analyze the causes and implications of climate variability and climate change in relation to human–environment systems and sustainable development.

PROVISIONAL

Mapping with CLOs with PLOs

CLO / PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	✓									✓
CLO2	✓	✓		✓	✓					✓
CLO3	✓	✓	✓	✓	✓	✓			✓	✓
CLO4	✓	✓	✓	✓		✓				✓
CLO5	✓	✓		✓	✓	✓			✓	✓
CLO6	✓	✓	✓	✓	✓	✓	✓		✓	✓

Course Contents

Module No.	Contents	Teaching–Learning Strategies	Assessment Strategies	CLOs
1	**Atmospheric Science:** definition, scope and sub-classifications, history of atmospheric science, nature, origin and composition of the atmosphere, vertical divisions of the atmosphere; meteorological equipment and weather forecasting: surface measurements, application of remote sensing, satellite observation.	Lecture (PPT), short videos, group discussion, Q&A	Short quiz, class participation, assignment, identification of meteorological instruments	CLO1, CLO2
2	**Earth's Surface and Atmosphere:** solar insolation of the earth and heat budget	Lecture (PPT), short videos, group discussion, visual quiz	Visual quiz, numerical/problem-solving exercise, short assignment, class discussion	CLO1, CLO2
3	**Water Balance of the Atmosphere:** humidity, physical changes of state of water, processes of cooling to produce condensation and sublimation, cloud formation and classification, fog formation and types, precipitation, causes, forms and types.	Lecture (PPT), short videos, group discussion, case-based learning, Q&A	Quiz, diagram-based assessment, case-study analysis, short assignment	CLO2
4	**Air Pressure and Winds:** pressure, vertical distribution of pressure, pressure gradient and wind, factors affecting wind speed and direction; general atmospheric circulation, seasonal changes in general circulation, oceanic circulation, monsoon,	3D globe activity, interactive lecture, visual quiz, group discussion, Q&A	Visual quiz, map/diagram interpretation, case-study analysis, written assignment, class presentation	CLO3, CLO5, CLO6

PROVISIONAL

	local winds, mountain and valley breeze; stability and instability, air masses, source regions and classification, extratropical cyclones, anticyclones, thunderstorms.			
5	**Climate Controls and Climate:** definition, production of climates, climatic classification of the world and world's climatic regions, role of oceans, *El Niño–Southern Oscillation (ENSO)* events, *La Niña* events, preliminary concepts of climate change.	Interactive lecture, group discussion, case-study analysis on human–environment and climate, Q&A	Climate-data interpretation, case-study report, quiz, presentation, written assignment	CLO4, CLO5, CLO6

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Ahmed, R. (2015). *Meteorology and climate change* (5th ed.). Sujoneshu Publications.

Barry, R. G., & Chorley, R. J. (2009). *Atmosphere, weather and climate* (1st ed.). Routledge.

Cole, F. W. (1980). *Introduction to meteorology* (3rd ed.). John Wiley & Sons.

Edward, B. (1997). *Climate process and change*. Cambridge University Press.

IPCC. (2023). *Climate change 2023: Synthesis report*. Intergovernmental Panel on Climate Change.

Trewartha, G. T., & Horn, L. H. (2017). *An introduction to climate* (5th ed.). Routledge

PROVISIONAL

Course Code	8224455	Marks: 100	Credits: 4	Class Hours: 56
Course Title	Physical Geography and Environment: Bangladesh Perspective			

Course Description

This course provides an integrated understanding of the geological, geomorphological, climatic, hydrological, and pedological characteristics of Bangladesh. It examines the geological evolution and landform development of the country, including the Ganges-Brahmaputra-Meghna delta, physiographic regions, riverine and coastal environments, estuaries, islands, beaches, dunes, tidal flats, and offshore features. Emphasis is placed on the interactions among geological processes, landforms, climate, hydrology, soils, and environmental change, with particular attention to the physical diversity and environmental characteristics of Bangladesh.

Course Objectives

By the end of the course, students will be expected to:

1. Develop a comprehensive understanding of the geological origin, evolution, and physical development of Bangladesh.
2. Explain the relationships among geology, geomorphology, climate, hydrology, soil formation, and landscape development in Bangladesh.
3. Develop the ability to identify, describe, classify, and interpret the major physiographic, topographic, riverine, estuarine, coastal, and offshore features of Bangladesh.
4. Analyze the spatial variations of physical environments and the processes responsible for landscape development and environmental change across Bangladesh.

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome
CLO1	Explain the geological origin, evolution, and development of the present physical structure of Bangladesh.
CLO2	Analyze the relationships between geological history, geomorphological processes, and the development of present-day landforms in Bangladesh.
CLO3	Describe and interpret the major characteristics and spatial variations of climate, hydrology, and soil formation across Bangladesh.
CLO4	Identify, classify, and interpret the major physiographic regions, sub-regions, topographic features, and environmental variations of Bangladesh.
CLO5	Explain and interpret the formation and characteristics of riverine, estuarine, coastal, and offshore landforms and associated depositional processes.
CLO6	Apply geographical knowledge and map-based analysis to interpret the spatial distribution, diversity, and environmental characteristics of Bangladesh's physical landscapes.

PROVISIONAL

Mapping with CLOs with PLOs

CLO / PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	√	√								√
CLO2	√	√	√	√						√
CLO3	√	√	√	√		√				√
CLO4	√	√	√			√				√
CLO5	√	√	√	√		√				√
CLO6	√	√	√	√	√	√			√	√

Course Content

Module No.	Content	Teaching–Learning Strategies	Assessment Strategies	CLOs
01	**Introduction:** Physical geography and environmental connections; scope, location and extent of Bangladesh; regional context and geographical significance.	Lecture (PPT), group discussion, case-based learning, Q&A	Short quiz, class participation, short assignment, concept-mapping exercise	CLO1, CLO4
02	**Geology:** Regional geological setting; geological structure; geological succession; stratigraphy and structural features; tectonic evolution; earthquakes; composition and characteristics of sediments.	Lecture with visual presentation, group discussion, case-based learning, Q&A	Quiz, geological diagram interpretation, short assignment, case-study analysis	CLO1, CLO2
03	**Geomorphology:** Overview of geomorphology; history and development of the delta; floodplains; uplifted blocks; hills; river changes; Ganges-Brahmaputra-Meghna Delta.	Interactive lecture, short videos, group discussion, case-study analysis, Q&A	Quiz, geomorphological map interpretation, case-study report, class presentation	CLO1, CLO2, CLO5
04	**Climate:** Characteristics of climate; seasons; climatic parameters; climate variability and climate change.	3D globe activity, interactive lecture, visual quiz, group discussion, Q&A	Visual quiz, climate-data interpretation, short assignment, case-study analysis	CLO3
05	**Hydrology:** Rivers; flood and flooding processes; groundwater and groundwater resources.	Lecture (PPT), group discussion, short videos	Quiz, river/flood map interpretation, case-study analysis, written assignment	CLO3, CLO4

PROVISIONAL

06	**Soils:** Soil formation; diversity and complexity of soils; soil classification and mapping; soil description and characteristics.	Lecture (PPT), soil map and chart discussion, Q&A	Quiz, soil-map interpretation, soil-profile exercise, short assignment	CLO3, CLO4
07	**Physiographic Regions and Sub-regions:** Major physiographic regions and sub-regions; characteristics, distribution, relief, and environmental significance.	Lecture (PPT), map demonstration, group discussion, case-study analysis, Q&A	Physiographic map exercise, map interpretation, quiz, case-study assignment	CLO4, CLO6
08	**Estuarine Environment:** Occurrence and distribution; sources and nature of estuarine sediments; estuarine processes; hydrodynamic factors; sediment dispersal and depositional systems; depositional sequences in estuaries.	Interactive lecture, visual quiz, group discussion, Q&A	Quiz, estuarine process diagram, case-study analysis, short presentation	CLO2, CLO3, CLO5
09	**Inland Riverine Sandbars and Offshore Islands:** Formation, distribution, morphology, sedimentation, and environmental characteristics of riverine sandbars and offshore islands.	Lecture with visual presentation, group discussion, case-based learning, Q&A	Map-based exercise, visual identification test, case-study assignment, quiz	CLO2, CLO5, CLO6
10	**Beaches, Dunes, Tidal Mudflats and Marshes:** Formation, distribution, morphology, sediment characteristics, depositional processes, and environmental significance.	Lecture (PPT), group discussion, short videos, visual presentation	Visual identification, diagram interpretation, short assignment, case-study analysis	CLO2, CLO5, CLO6
11	**Bengal Sea Fan and Swatch of No Ground:** Geological and geomorphological characteristics; sedimentation; submarine landforms; environmental and geographical significance.	Lecture with visual presentation, group discussion, case-based learning, Q&A	Quiz, map interpretation, short assignment, case-study presentation	CLO1, CLO2, CLO5, CLO6

PROVISIONAL

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Brammer, H. (2012). *The physical geography of Bangladesh*. The University Press Limited.

Khan, F. H. (1991). *Geology of Bangladesh*. The University Press Limited.

Rasheed, K. B. S. (2016). *Bangladesh: Resource and environmental profile* (2nd ed.). A. H. Development Publishing House.

Van Schendel, W. (2020). *A history of Bangladesh* (2nd ed.). Cambridge University Press.

PROVISIONAL

Course Code	8224456	Marks: 100	Credits: 4	Class Hours:56
Course Title	Environment and Economics			

Course Description

This course introduces the fundamental principles of environmental and natural resource economics and their application to environmental management and sustainable development. It examines the economic relationships between the environment, natural resources, markets, population, and economic activities. The course covers environmental valuation, externalities, cost benefit and cost-effectiveness analysis, pollution control, population dynamics, ecosystem services, carbon taxation, and the blue economy, with emphasis on applying economic approaches to contemporary environmental challenges and policy decisions.

Course Objectives

By the end of the course, students will be expected to:

1. Develop an understanding of the fundamental concepts, principles, and scope of environmental and natural resource economics and their relationship with environmental management.
2. Apply basic economic theories, models, and analytical tools to environmental and natural resource problems, including externalities, pollution, resource allocation, and environmental quality.
3. Analyze population dynamics, population environment relationships, and their implications for natural resource use and sustainable development.
4. Evaluate environmental policies and management options using economic approaches such as environmental valuation, cost-benefit analysis, cost-effectiveness analysis, carbon taxation, and ecosystem-service valuation.

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome
CLO1	Explain the fundamental concepts, principles, scope, and interdisciplinary relationships of environmental and natural resource economics.
CLO2	Analyze the economic relationships among markets, natural resources, blue economies, environmental quality, pollution, and ecosystem services.
CLO3	Apply basic economic concepts and analytical tools, including supply and demand, externalities, willingness to pay, efficiency, and pollution-control models, to environmental problems.
CLO4	Analyze population dynamics, including fertility, mortality, migration, population growth theories, and population–environment relationships.
CLO5	Evaluate environmental management alternatives using cost–benefit analysis, cost-effectiveness analysis, environmental valuation, and risk/damage assessment.
CLO6	Interpret and recommend economically informed approaches to contemporary environmental challenges, including ecosystem-service management, carbon taxation, carbon footprints, and the blue economy.

PROVISIONAL

Mapping with CLOs with PLOs

CLO / PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	√	√				√				√
CLO2	√	√	√	√		√	√			√
CLO3	√	√	√	√		√				√
CLO4	√	√		√		√				√
CLO5	√	√	√	√		√	√		√	√
CLO6	√	√	√	√		√	√		√	√

Course Content

Module No.	Content	Teaching–Learning Strategies	Assessment Strategies	CLOs
01	**Environmental Economics:** Definition, scope and goals; economic analysis; incentives in households, industry and transportation; environmental policy design; environment and economic growth; cost-effectiveness analysis; benefit-cost analysis; international environmental issues; economics and politics.	Lecture (PPT), peer-feedback discussion, case-based learning, Q&A	Short quiz, concept-mapping exercise, case-study analysis, short assignment	CLO1 CLO2 CLO5
02	**The Economy and Environment:** Natural resource economics; fundamental economic-environmental balance; environment as an economic and social asset; GDP and environmental indicators; emissions, ambient quality and damages; types of pollutants; short-run and long-run choices.	Lecture with visual presentation, group discussion, case-based learning	Quiz, case-study analysis, written assignment, class discussion	CLO1 CLO2
03	**Analytical Tools:** Benefits and costs; supply and demand; willingness to pay; aggregate demand; benefits, costs and technology; economic efficiency and markets; efficiency and equity; external costs and benefits; environmental quality; pollution-control models; pollution damages; abatement costs; efficient level of emissions;	Lecture (PPT), diagram-based discussion, case-based learning, problem-solving exercises, Q&A	Numerical/problem-solving exercises, quiz, diagram interpretation, case-study assignment	CLO2 CLO3

PROVISIONAL

	enforcement costs; equimarginal principle; long-run sustainability.			
04	**Environmental Analysis:** Analytical framework; impact analysis; cost-effectiveness analysis; damage assessment; benefit–cost analysis; risk analysis; environmental benefits and costs.	Lecture (PPT), diagram-based discussion, case-based learning, Q&A	Cost–benefit exercise, case-study analysis, written assignment, class presentation	CLO3 CLO5
05	**Population Dynamics:** Fertility measures and determinants; economic model of family size; fertility trends in Bangladesh; mortality measures, determinants and consequences.	Lecture (PPT), peer-feedback discussion, diagram-based discussion, case-based learning, Q&A	Quiz, demographic data interpretation, short assignment, case-study analysis	CLO3 CLO4
06	**Theories of Population Growth:** Malthusian and optimum population concepts; demographic transition theory; theories and models of migration; Harris–Todaro model; population and environment; population policies and programmes in Bangladesh.	Lecture (PPT), peer-feedback discussion, case-based learning, model interpretation, Q&A	Theory-based quiz, model interpretation, case-study report, presentation	CLO4 CLO6
07	**Ecosystem Services:** Types, characteristics, functions, economic importance, and valuation of ecosystem services.	Interactive lecture, group discussion, short videos, visual presentation, Q&A	Ecosystem-service valuation exercise, case-study analysis, group presentation, short assignment	CLO2 CLO5 CLO6
08	**Blue Economy, Carbon Tax and Carbon Footprint:** Concept and components of the blue economy; economic opportunities and environmental sustainability; carbon tax; carbon footprint; economic instruments for climate and environmental management.	Lecture (PPT), group discussion, visual presentation, case-based learning, Q&A	Carbon-footprint calculation, policy case study, group presentation, written assignment	CLO2 CLO5 CLO6

PROVISIONAL

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Field, B. C. (1997). *Environmental economics: An introduction* (2nd ed.). McGraw-Hill.

Hauser, P. M., Duncan, O. D., & Dudley, D. (1972). *The study of population: An inventory and appraisal*. University of Chicago Press.

Wonnacott, P., & Wonnacott, R. (1986). *Economics* (3rd ed.). McGraw-Hill.

Course Code	8224457	Marks: 100	Credits: 4	Class Hours:56
Course Title	Environmental Biochemistry			

Course Description

This course provides a fundamental understanding of biochemical principles and processes operating within environmental systems, with particular emphasis on soil, water, living organisms, and environmental pollutants. It covers cell structure and division, biomolecules, vitamins and minerals, metabolism, xenobiotic degradation, enzymatic processes, metal toxicity, cellular injury, and biotic and abiotic biochemical interactions. The course enables students to understand how biochemical processes influence environmental quality, organismal health, pollutant transformation, and ecosystem functioning

Course Objectives

The course aims to:

1. Develop a fundamental understanding of biochemical principles and their interactions within soil, water, organisms, and environmental systems.
2. Explain the structure and functions of cells, sub-cellular organelles, and cell division, and relate these processes to environmental and biological functions.
3. Develop knowledge of biomolecules, vitamins, minerals, and their biochemical roles in living organisms and environmental systems.

PROVISIONAL

4. Explain biochemical metabolism, degradation of environmental contaminants, toxicity, and mechanisms of cellular and tissue damage.
5. Develop an understanding of biotic and abiotic biochemical reactions and their significance in environmental processes and environmental health.

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome
CLO1	Explain the fundamental principles, historical development, scope, and applications of biochemistry in environmental science.
CLO2	Explain cell structure, functions of sub-cellular organelles, cellular organization, and the processes of mitosis and meiosis, including their environmental and biological significance.
CLO3	Describe the structures, classification, properties, functions, and biological significance of carbohydrates, lipids, amino acids, peptides, proteins, vitamins, and minerals in environmental and biological systems.
CLO4	Explain metabolic processes, xenobiotic metabolism, enzymatic degradation, cytochrome P450-mediated reactions, free-radical formation, lipid peroxidation, and biochemical mechanisms of cellular and tissue injury.
CLO5	Analyze biotic and abiotic biochemical interactions, metal–biomacromolecule interactions, metal toxicity, and biochemical processes influencing environmental quality and living systems.

Mapping with CLOs with PLOs

CLO / PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	✓					✓				✓
CLO2	✓	✓	✓			✓				✓
CLO3	✓	✓	✓			✓				✓
CLO4	✓	✓	✓	✓		✓				✓
CLO5	✓	✓	✓	✓		✓	✓			✓

Course Contents

Module No.	Content	Teaching–Learning Strategies	Assessment Methods	CLOs
1	**History, Scope and Future of Biochemistry:** Historical development, scope, future directions, and appreciation of biochemistry in environmental science.	Interactive lecture, PPT presentation, group discussion, question-and-answer	Class participation, quiz, short-answer questions	**CLO1**
2	**The Cell and Its Evolution:** Cell evolution, structure and functions of sub-	Lecture, PPT, short videos, group	Quiz, class test, short-	**CLO2**

PROVISIONAL

	cellular organelles, isolation and identification of organelles, and brief treatment of meiosis and mitosis.	discussion, case-based learning, Q&A	answer questions	
3	**Biomolecules: Carbohydrates:** Biological functions, classification and nomenclature, optical properties, ring structure of common monosaccharides, proof of ring structure of glucose, mutarotation, general properties and colour tests of reducing sugars, derivatives of monosaccharides, and sugar acids. **Disaccharides:** Maltose, lactose, sucrose and other disaccharides, isolation from natural sources, structures and biological importance. **Polysaccharides:** Storage and structural polysaccharides; structures and functions of starch, glycogen, cellulose, and other biologically important polysaccharides. **Lipids:** Nomenclature, classification, general reactions of fats and fatty acids, sterols, structures and biological functions of different classes of lipids, and isolation of cholesterol and phospholipids from natural sources. **Amino Acids and Peptides:** Structural features, optical activity, classification, and physicochemical properties. **Proteins:** General introduction, classification based on biological functions, shape and structure, isolation and purification, primary structure, sequence determination of insulin, sequence homology, and denaturation.	Interactive lecture, PPT, molecular/structural representation, group discussion, concept mapping, problem-based learning, Q&A	Written examination, class test, structural questions, assignment, short-answer questions	**CLO3, CLO5**
4	**Vitamins and Minerals:** Occurrence, classification, physical and chemical properties, physiological functions, deficiency symptoms; metal toxicity; chemical forms; metal–biomacromolecule interactions; toxic burden from the environment; and role of metals in metabolism.	Lecture, PPT, case-based learning, group discussion, concept mapping, Q&A	Quiz, assignment, case analysis, written examination	**CLO3, CLO5**

PROVISIONAL

5	**Metabolism:** Anabolism and catabolism; central metabolic pathways of proteins, lipids and carbohydrates; xenobiotic metabolism; enzymatic degradation by monooxygenases; role of cytochrome P450 and its multiple forms; cellular and tissue injury; altered membrane permeability; free-radical formation; lipid peroxidation; lysosomal degradation; and superoxide dismutase.	Interactive lecture, PPT, concept mapping, pathway diagrams, case-based learning, problem-solving, Q&A	Written examination, concept-map assessment, class test, case-based questions	**CLO3, CLO4, CLO5**

Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Berg, J. M., Tymoczko, J. L., Gatto, G. J., Jr., & Stryer, L. (2019). *Biochemistry* (9th ed.). W. H. Freeman and Company.

Karp, G. (2013). *Cell and molecular biology: Concepts and experiments* (7th ed.). John Wiley & Sons.

Nelson, D. L., & Cox, M. M. (2021). *Lehninger principles of biochemistry* (8th ed.). W. H. Freeman and Company.

Voet, D., Voet, J. G., & Pratt, C. W. (2016). *Fundamentals of biochemistry: Life at the molecular level* (5th ed.). John Wiley & Sons.

PROVISIONAL

Course Code	8224460	Marks: 100	Credits: 4	Class Hours:56
Course Title	Laboratory and Field Work on Environment Science			

Course Description

This practical course develops laboratory and field-based skills in environmental chemistry, earth and soil sciences, and environmental biology. Students gain hands-on experience in environmental sampling, water and soil analysis, geological and biological specimen identification, ecological surveys, data interpretation, and scientific reporting

Course Objectives

The course aims to:

1. Develop practical skills in laboratory analysis of water, wastewater, soil, rocks, minerals, and biological specimens.
2. Develop field skills in environmental observation, sampling, measurement, specimen collection, and ecological investigation.
3. Enable students to apply appropriate laboratory and field techniques for analyzing and interpreting environmental conditions.
4. Develop competency in scientific data recording, analysis, documentation, and communication through laboratory notebooks, field reports, and practical presentations.

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome
CLO1	Perform basic laboratory procedures for the preparation, standardization, and quantitative analysis of environmental samples using appropriate techniques and instruments.
CLO2	Measure and interpret major physical and chemical parameters of water and wastewater and assess their environmental significance.
CLO3	Collect, characterize, and analyze soil, rock, and mineral samples using appropriate field and laboratory methods.
CLO4	Identify and characterize selected biological, geological, and environmental specimens using appropriate identification techniques.
CLO5	Apply field-based methods to investigate environmental conditions, ecosystem composition, plant populations, and biological characteristics.
CLO6	Analyze, document, and communicate laboratory and field observations through scientific notebooks, data tables, calculations, practical reports, and field reports.

PROVISIONAL

Mapping with CLOs with PLOs

CLO / PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	✓	✓		✓	✓		✓		✓	
CLO2	✓	✓	✓	✓	✓	✓	✓		✓	✓
CLO3	✓	✓		✓	✓		✓		✓	
CLO4	✓	✓		✓	✓		✓		✓	
CLO5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CLO6	✓	✓		✓			✓	✓	✓	✓

Course Content

A. Environmental Chemistry

Practical No.	Practical Content	Teaching–Learning Strategies	Assessment Strategies	CLOs
C1	Preparation and standardization of acid and base solutions.	Laboratory demonstration, instructor-guided practice, hands-on titration, safety briefing, Q&A.	Practical performance, calculation, laboratory notebook, viva voce.	CLO1 CLO6
C2	Determination of pH, EC, turbidity, TDS, TSS, total solids, total hardness, total alkalinity, total acidity, DO, BOD, and COD of water and wastewater.	Instrument demonstration, laboratory experimentation, sample preparation, group practice, data recording and interpretation.	Practical test, analytical calculations, data interpretation, laboratory report, viva voce.	CLO1 CLO2 CLO6
C3	Determination of Fe, Cu, Pb, Cr, Ni, Ca, and Al in water and wastewater.	Laboratory demonstration, instrumental analysis, guided experimentation, sample analysis, data interpretation.	Practical performance, analytical calculations, result interpretation, laboratory report, viva voce.	CLO1 CLO2 CLO6
C4	Complexometric titration using EDTA with Murexide and Eriochrome Black T/Solochrome Black indicators.	Demonstration, step-by-step laboratory practice, guided titration, peer learning, Q&A.	Titration performance, calculation, practical notebook, viva voce.	CLO1 CLO6
C5	Gravimetric determination of chloride, carbonate, sulphate, and phosphate.	Laboratory demonstration, hands-on experimentation, sample preparation, guided observation, discussion.	Practical performance, calculation, laboratory report, observation of technique, viva voce.	CLO1 CLO2 CLO6

PROVISIONAL

B. Earth and Soil Sciences

Practical No.	Practical Content	Teaching–Learning Strategies	Assessment Strategies	CLOs
E1	Field visits to environmentally important areas.	Field-based learning, direct observation, instructor demonstration, group investigation, field discussion.	Field participation, observation sheet, field notebook, field report.	CLO3 CLO5 CLO6
E2	Description and identification of minerals in outcrops.	Field demonstration, specimen observation, comparative identification, guided discussion.	Specimen identification test, field notebook, oral questioning.	CLO3 CLO4
E3	Investigation of soil conditions in selected areas.	Field observation, soil-profile examination, field measurement, group investigation.	Field observation report, soil description exercise, practical assessment.	CLO3 CLO5 CLO6
E4	Collection, labeling, preservation, and documentation of soil and rock samples.	Field demonstration, hands-on sampling, sample-labeling practice, safety and preservation techniques.	Sampling technique, sample labeling, field notebook, instructor observation.	CLO3 CLO6
E5	Laboratory analysis and interpretation of collected soil and rock samples.	Laboratory demonstration, hands-on analysis, microscopy, guided interpretation, group discussion.	Laboratory performance, analytical report, specimen interpretation, viva voce.	CLO3 CLO4 CLO6
E6	Preparation and submission of field-tour report.	Field documentation, scientific report-writing guidance, data organization, peer feedback.	Field report, data presentation, interpretation, scientific writing.	CLO5 CLO6
E7	Determination of soil texture by particle-size analysis using the hydrometer method.	Laboratory demonstration, hands-on experiment, calculation exercise, guided data interpretation.	Practical performance, calculation, texture-classification exercise, laboratory report.	CLO3 CLO6
E8	Determination of soil bulk density and particle density.	Demonstration, hands-on measurement, calculation practice, discussion of results.	Practical test, numerical calculation, result interpretation, laboratory notebook.	CLO3 CLO6
E9	Identification of common igneous, sedimentary, and	Microscopic demonstration, guided	Thin-section identification test,	CLO3 CLO4

PROVISIONAL

	metamorphic rocks in thin section.	slide observation, comparative identification, discussion.	practical notebook, viva voce.	

C. Environmental Biology

Practical No.	Practical Content	Teaching–Learning Strategies	Assessment Strategies	CLOs
B1	Field survey on Environmental Biology.	Field observation, ecological survey, group investigation, instructor guidance, field discussion.	Field participation, observation sheet, field notebook, survey report.	CLO4 CLO5 CLO6
B2	Collection techniques for phytoplankton and zooplankton.	Demonstration, hands-on sampling, specimen handling, preservation techniques, group practice.	Collection technique, specimen handling, practical observation, viva voce.	CLO4 CLO5
B3	Identification and characterization of phytoplankton and zooplankton.	Microscopic observation, taxonomic keys, specimen comparison, guided identification, individual practice.	Specimen identification test, practical examination, identification record, viva voce.	CLO4 CLO6
B4	Study of adaptive characteristics of hydrophytes, xerophytes, mesophytes, and halophytes.	Field observation, specimen examination, microscopy, comparative analysis, group discussion.	Adaptation-identification exercise, comparative assignment, practical viva.	CLO4 CLO5
B5	Study of reproductive behavior of selected plant and animal species.	Field observation, video-based learning, behavioral recording, case-based discussion.	Observation report, behavioral interpretation, presentation, viva voce.	CLO4 CLO5 CLO6
B6	Study of plant population using the quadrat method.	Field demonstration, quadrat sampling, hands-on data collection, numerical analysis, group work.	Field performance, population-data calculation, field report, practical assessment.	CLO5 CLO6
B7	Study of ecosystem composition in selected areas.	Field survey, ecosystem observation, species/community documentation, group	Ecosystem survey report, data interpretation, presentation, field	CLO4 CLO5 CLO6

PROVISIONAL

		investigation, discussion.	notebook.	
B8	Determination of dissolved oxygen (DO).	Laboratory demonstration, water-sample preparation, hands-on analysis, calculation and interpretation.	Practical performance, calculation, laboratory report, viva voce.	CLO1 CLO2 CLO6
B9	Preparation of permanent slides.	Laboratory demonstration, specimen preparation, microscopy, individual practice, instructor feedback.	Slide-preparation technique, microscopic observation, practical examination.	CLO4 CLO6
B10	Preparation and maintenance of field and laboratory notebooks.	Demonstration of scientific documentation, guided recording, data organization, instructor feedback.	Notebook assessment, completeness, accuracy of observations, presentation and documentation quality.	CLO6

Course Assessment Framework

Total Course Marks: 100 Examination Duration: **Time: 6 (Six) Hours**

Distribution of Marks for 2nd year Final Examination (Section A, B & C)

1. **Final Exam: 60 marks**

Section A : Environmental Chemistry: 18 marks

a) Five questions shall be set from the five modules, of which candidates shall answer any three (3x6=18)

b) Each question carries 6 marks

c) Each question will be segmented into (a/b/c) or (a/b) as applicable.

Section B : Earth and Soil Science: 24 marks

a) Five questions shall be set from the selected modules (E3, E4, E5, E7, E8 & E9) of which candidates shall answer any three (3x8=24)

PROVISIONAL

b) Each question carries 8 marks
c) Each question will be segmented into (a/b/c) or (a/b) as applicable.
d) From E5 & E9, Questions may include definition, origin and occurrence, physical properties, mineral hardness and the Mohs hardness scale, classification, comparison, specimen identification and environmental applications of rocks and minerals, as applicable.
e) From E3, E4, E7 & E8, Questions may include definition, classification, comparison, Marshal's texture triangle, profile characteristics and their importance, significance, as applicable.

Section C: Environmental Biology: 18 marks

a) Five questions shall be set from the selected (B2, B3, B4, B5, B6, B7, B8 & B9) modules, of which candidates shall answer any three (3x6=18)
b) Each question carries 6 marks
c) Each question will be segmented into (a/b/c) or (a/b) as applicable.
d) From B2, B3, B4, B5, B6, B7, B8 & B9, Questions may include definition, characteristics, examples, collection techniques, procedure as applicable.

2. **Lab Notebook: 10 Marks**
3. **Lab Viva-voce: 10 Marks**
4. **Presentation: 10 Marks**
5. **Field Report: 10 Marks (Day-long fieldwork)**

PROVISIONAL

Guidelines of Viva-voce

Course Code: 8224464	**Marks:** 50	**Credits:** 2
Course Title: Viva-voce (Environmental Science)	**Paper Orientation:** Practical	

Course Description

The Viva-Voce is a comprehensive oral examination designed to evaluate students' overall academic achievement, professional competence, critical thinking, and communication skills acquired during the B.Sc. (Honours) in Environmental Science program. Conducted at the end of the program after the completion of written and practical examinations, the course provides students with an opportunity to demonstrate their understanding of core environmental concepts, ecology, climate change, biodiversity, pollution control, and natural resource management. The examination also assesses students' ability to integrate interdisciplinary scientific knowledge, apply modern geospatial and analytical tools, solve real-world environmental problems, and communicate scientific findings effectively in academic and professional settings.

Course Objectives

The course's objectives are to:

- Evaluate students' comprehensive scientific knowledge of environmental science principles, ecology, pollution, climate systems, and natural resource management.
- Assess students' ability to apply interdisciplinary knowledge and modern tools (such as GIS and remote sensing) in addressing environmental monitoring, assessment, and problem-solving.
- Develop effective oral communication, technical presentation, and scientific reporting skills for diverse audiences.
- Evaluate students' critical thinking, data analysis, and evidence-based decision-making abilities regarding ecological conservation and sustainable development goals (SDGs).
- Assess students' professional ethics, environmental responsibility, leadership qualities, and collaborative teamwork skills.
- Prepare students for professional careers in environmental consulting, research institutions, government and non-government organizations, and advanced academic studies.

Course Learning Outcomes (CLOs)

After going through the viva voce, the students will be able to:

CLO 1: Demonstrate comprehensive knowledge of core environmental concepts, including ecology, climate, biodiversity, pollution, and natural resource management.

CLO 2: Apply scientific principles and modern technological tools (e.g., GIS, remote sensing, analytical models) to evaluate and solve environmental challenges.

CLO 3: Design scientific investigations, collect and interpret environmental field/laboratory data, and formulate sustainable, evidence-based solutions.

CLO 4: Communicate scientific ideas, research outcomes, and environmental policies clearly and professionally through oral presentations and technical reports.

PROVISIONAL

CLO 5: Exhibit strong ethical principles, environmental responsibility, leadership, and effective teamwork in managing multidisciplinary ecological projects.

CLO 6: Demonstrate career readiness, adaptability, and a commitment to continuous professional development and lifelong learning in environmental fields.

CLO to PLO Mapping Matrix

CLO / PLO	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10
CLO 1	✓	✓	-	-	-	✓	-	-	-	-
CLO 2	✓	✓	✓	✓	✓	✓	-	-	-	-
CLO 3	-	✓	✓	✓	✓	✓	✓	-	-	-
CLO 4	-	-	-	-	-	-	-	✓	-	✓
CLO 5	-	-	-	-	-	-	✓	✓	✓	-
CLO 6	-	-	-	-	-	-	-	-	-	✓

Note: (✓) = Alignment/Correlation exists; (-) = No Correlation.

Teaching System

The course is conducted through an interactive and student-centered approach. Teaching and learning activities include orientation sessions, guided discussions, question-and-answer sessions, seminar presentations, mock viva examinations, field and laboratory project reviews, individual consultations, self-directed learning, and revision of core Environmental Science concepts. Students are encouraged to develop sound communication, critical thinking, analytical, and presentation skills in preparation for the final viva voce examination.

Assessment Strategy

Students will be assessed through a comprehensive viva voce examination conducted by the designated Examination Committee following the completion of written and practical examinations. The assessment will evaluate students' overall conceptual understanding of environmental science, their ability to integrate theoretical knowledge with field and technological applications, critical thinking and problem-solving capacities, oral communication skills, professional attitude, environmental stewardship, and responsiveness to technical questions. The final grade will be awarded based on the comprehensive performance demonstrated during the viva voce examination.

PROVISIONAL